A smart mushroom incubation system for enhanced yield and sustainability in small scale farming

RP 25-26J-264

Project Proposal Report

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BSc (Hons) in Information Technology Specializing in Information Technology

Department of Information Technology
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Declaration

We declare that this is our own work, and this proposal does not incorporate without acknowledgement any material previously submitted for a degree or diploma in any other university or Institute of higher learning and to the best of our knowledge and belief it does not contain any material previously published or written by another person except where the acknowledgement is made in the text.

29/08/2025

Signature Date

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Signature of the Supervisor Date

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Signature of the Co-Supervisor Date

Abstract

The purpose of this research is to improve the reliability and sustainability of oyster mushroom farming in Sri Lanka through the introduction of a Smart Mushroom Incubation System [1][2]. Mushroom cultivation is widely practiced across the country, particularly by small-scale farmers, because of its relatively low initial investment and high nutritional value [3][4]. However, the industry continues to face several persistent challenges that reduce both yield and quality. Manual monitoring of environmental conditions requires constant attention and is often inaccurate, leading to unfavorable growth environments [5]. In addition, inconsistent yields from one cycle to another, combined with the risk of pest infestations, result in financial losses and discourage many farmers from expanding their operations [6]. These problems highlight the need for a modern, technology-driven solution that reduces labor demands while improving efficiency and crop health [7].

To address these issues, this study proposes the design and development of an IoT-based Smart Incubation System tailored for oyster mushroom cultivation [8]. The system integrates a network of sensors to continuously monitor critical environmental factors such as temperature, humidity, and carbon dioxide concentration [9]. Data is collected in real-time and made accessible through a mobile-friendly dashboard, which provides farmers with actionable insights and instant alerts when conditions move beyond the ideal range. By enabling early identification of unfavorable changes and potential pest risks, the system allows farmers to respond quickly and maintain stable growing conditions without relying on constant manual observation [10].

The findings suggest several important benefits. Automated monitoring reduces dependence on manual labor, saving farmers time and effort while lowering the chances of human error [2]. More consistent environmental control contributes to steady yields and improved crop quality, helping farmers secure better prices in the market [3]. Additionally, by alerting farmers to potential risks at an early stage, the system minimizes crop losses due to pests and reduces the unnecessary use of chemical pesticides, supporting healthier and more sustainable farming practices [6][7].

In summary, the Smart Mushroom Incubation System represents a practical, low-cost solution for improving the efficiency and profitability of mushroom farming in Sri Lanka [1]. By combining traditional farming knowledge with modern IoT technology, this research demonstrates how small-scale farmers can enhance productivity, reduce risks, and move toward more sustainable agricultural practices [4][5]. In the long term, the adoption of such systems has the potential to transform mushroom cultivation into a more reliable, technology-driven industry that strengthens rural livelihoods and contributes to food security in the region [9][10].

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Introduction

Mushroom farming is becoming an important part of agriculture in Sri Lanka, especially for small farmers who depend on it for both income and food [1]. One of the biggest problems they face is fungal contamination, such as green mold or cobweb disease [2]. These infections spread very quickly and can ruin a large part of the harvest [3]. Now, farmers must check each mushroom packet by hand every day. This takes a lot of time, it is hard to manage when there are many packets, and sometimes the farmer misses the early signs of contamination [4]. Even worse, touching the packets too often can spread the fungus from one to another [5]. Because of this, many farmers end up with big losses that could have been avoided with early detection [6].

Over the years, researchers have tried different ways to solve similar problems in agriculture. Techniques like computer vision and artificial intelligence (AI) have been used to detect diseases in crops by analyzing images of plant leaves [7]. Deep learning models, especially Convolutional Neural Networks (CNNs), have shown very good results in identifying patterns of disease [8]. Some studies used fixed cameras in mushroom farms to spot mold, while others-built datasets of healthy and infected crops under control conditions and trained AI models to classify them [9]. These methods worked well in experiments, often reaching very high accuracy [10].

However, most of these solutions have limitations. Fixed cameras cannot cover all mushroom packets, and building datasets in labs does not match the messy, real-world conditions on a farm [3]. They are also often expensive and not designed for small farmers in places like Sri Lanka [6].

This research takes a different approach by using a mobile camera system with AI. Instead of staying fixed in one place, the camera moves along the mushroom rows and captures images of every packet [9]. A lightweight deep learning model, designed to run on mobile devices, then analyzes these images to detect early signs of fungal contamination [8]. Farmers get instant alerts on a simple dashboard and can even give feedback to help the system learn and improve [7]. This makes the solution affordable, easy to use, and better suited to the needs of small-scale mushroom farmers [1].

Background & Literature survey

Oyster mushroom farming has become a popular and profitable activity in Sri Lanka, especially for small-scale farmers. Mushrooms are easy to grow, have a short cultivation cycle, and provide both nutritional value and income for rural communities. Despite these advantages, mushroom cultivation faces significant challenges, the most critical being contamination. Fungal infections, such as green mold, and bacterial infections can spread rapidly among mushroom packets, destroying large portions of the crop. Early detection of contamination is essential because once it spreads, it becomes extremely difficult to control, leading to severe financial losses for farmers.

Traditionally, farmers monitor their crops through manual inspection, which involves checking each mushroom packet by hand. This process is time-consuming, labor-intensive, and often unreliable. Subtle signs of contamination are easy to lose, especially in larger cultivation setups. Frequent handling of mushroom packets can also unintentionally spread contamination from one packet to another. These challenges highlight the urgent need for an automated, accurate, and affordable solution for early detection of contamination.

The proposed research introduces a mobile camera system integrated with deep learning to address this problem. The camera moves row by row along the mushroom cultivation setup, capturing images of each packet for real-time analysis. A deep learning model processes these images to detect early signs of fungal or bacterial contamination and abnormal growth patterns. When contamination is detected, the system provides instant alerts to farmers through a simple, user-friendly dashboard, allowing immediate intervention. This automated approach reduces the need for manual inspections, improves detection accuracy, and ensures that no packet is missing during monitoring.

From a knowledge development perspective, this research combines agricultural practices with machine vision and artificial intelligence, offering a practical example of smart farming. Unlike traditional or lab-based methods, this system works in real cultivation conditions, capturing images under variable lighting and environmental factors. It also introduces a feedback loop where farmers can confirm or correct the system's alerts, enabling continuous learning and improvement of the detection model.

This research is important to society because it directly supports small-scale farmers, protecting their income and livelihood. Early detection of contamination reduces crop losses, saves resources such as water and substrate, and promotes sustainable farming practices. By providing an affordable and easy-to-use technology solution, the project empowers rural communities to adopt modern farming techniques, improve productivity, and contribute to food security. It also demonstrates how practical applications of AI and mobile imaging can benefit agriculture at the grassroots level, bridging the gap between technology and everyday farming needs.

Research Gap

Despite various studies in mushroom cultivation and disease detection, a notable gap exists in the development of systems that integrate real-time disease detection, automated health-based classification, and environmental monitoring. While previous research addressed individual aspects such as disease classification, environmental control, or general automation, no single study has combined all these features effectively. This absence is significant as it limits small- and medium-scale farmers from accessing a comprehensive, technology-driven solution for efficient and scalable mushroom farming. Thus, the development of an integrated system becomes vital to minimize crop losses, improve yield, and support sustainable mushroom cultivation practices.

Research A (AI-based Fruit Defect Detection)
This research presents a fully automated fruit grading system for apples, lemons, and oranges using SSD neural networks for surface defect detection. The system uses a dark box, ring-shaped LED lighting, and infrared sensors to capture images of fruits on a conveyor belt. It achieves high accuracy in detecting defects like bruises or pest damage. However, the study focuses only on fruit quality inspection and does not address automated cultivation, disease prevention, or real-time environmental control.

Research B (YOLOv5-based Mushroom Disease Detection)
This study employs YOLOv5 to detect fungal diseases in oyster mushrooms by classifying them as healthy or diseased. The system enables timely intervention to minimize crop loss and improve yield, demonstrating high precision and recall rates. Its limitation lies in focusing solely on disease classification, without integrating environmental monitoring or cultivation automation.

Research C (IoT- and ML-based Smart Mushroom Cultivation)
This research combines IoT and machine learning to automate mushroom cultivation, using ESP32 microcontrollers and sensors to monitor environmental factors such as temperature, humidity, and CO_2 levels. It also classifies mushroom types and separates diseased ones via image processing on a conveyor belt. While providing a holistic cultivation solution, the study emphasizes classification and general automation, rather than achieving highly accurate, real-time disease detection.

Research D (Mushroom Growth Optimization Kit)
This study introduces the Mushroom Growth Optimization (MGO) kit for small- and medium-scale mushroom farmers, automating the monitoring and adjustment of key environmental factors like temperature, humidity, and water levels. It focuses on sustainable, scalable cultivation and efficient resource use, but does not integrate advanced disease detection or automated classification of mushrooms based on health status.

Research E (Agrobot for Tomato Diseases and Pests)
This research develops a robotic system, Agrobot, using Raspberry Pi and YOLO models to detect tomato diseases and pests in real time. It navigates fields, scans crops, and classifies infections, aiming to reduce crop loss and improve precision in management. While it demonstrates effective disease detection in a different crop, it does not address integration with mushroom cultivation or real-time control of environmental conditions in indoor farming setups.

Application Reference	**Research A [1]**	**Research B [2]**	**Research C [3]**	**Research D [4]**	**Research E [5]**	**Proposed System**
Real-time disease detection	✗	✓	✗	✗	✓	✓
Environmental monitoring & control	✗	✗	✓	✓	✗	✓
Mushroom classification / type separation	✗	✗	✓	✗	✗	✓
Automation of cultivation	✗	✗	✓	✓	✗	✓
High accuracy detection	✓	✓	✗	✗	✓	✓

Research Problem

Fungal contamination is one of the most critical challenges in mushroom cultivation, particularly for small- and medium-scale farmers [1][2]. Early detection is difficult because manual inspection is slow, labor-intensive, and often fails to identify subtle signs such as mold, discoloration, or abnormal growth [3][4]. Existing AI-based systems are limited, as they are usually designed for fixed setups, are not mobile-friendly, or cannot adapt to new contamination patterns [5][6]. Moreover, current systems generally do not have the capability to scan and detect all mushroom packets row by row using a camera, requiring individual inspection or partial monitoring, which leaves gaps in early detection and reduces effectiveness [7][8].

The proposed Fungal Contamination Detection Module using Mobile Imaging and Deep Learning addresses these issues by enabling farmers to capture images of mushroom packets using a mobile device, scanning packets systematically row by row, and automatically detecting early signs of contamination through CNN-based analysis [9][10]. The system incorporates adaptive learning, allowing farmer feedback to retrain the model and improve accuracy over time [2][5]. Real-time alerts and a user-friendly mobile interface ensure that farmers can take immediate preventive action, reducing the spread of contamination and safeguarding crop quality [1][6].

The impact of this development is significant. It minimizes crop losses and improves overall yield by providing timely and accurate contamination detection [3][4]. It also reduces labor and inspection time, making mushroom farming more efficient and manageable [7][8]. By combining systematic camera-based scanning, real-time monitoring, deep learning, and adaptive learning, the system supports sustainable and scalable mushroom cultivation, offering farmers a reliable tool to maintain healthy crops and make informed decisions [9][10].

Objectives

In small-scale mushroom farming, farmers face the constant challenge of checking each mushroom packet by hand for signs of mold or fungal disease. This process is not only time-consuming but also increases the risk of spreading contamination through frequent handling. The aim of this research is to develop an automated system that can monitor mushroom packets continuously and accurately, using a combination of mobile imaging and deep learning. By scanning packets row by row, the system can detect early signs of fungal infection or abnormal growth patterns that might otherwise go unnoticed.

Once contamination is detected, the system sends real-time alerts to farmers through a simple and easy-to-use dashboard. This allows farmers to act quickly, isolating or treating affected packets before the problem spreads. The system also includes a feedback mechanism, so farmers can confirm or correct the detection results, helping the AI model to improve over time. Importantly, the solution is designed to be practical and cost-effective, ensuring it is accessible to small-scale farmers who may not have advanced technical resources. Overall, this research aims to provide a reliable, efficient, and user-friendly tool that reduces manual labor, protects crops, and supports sustainable mushroom farming practices.

Main Objectives

The main aim of this research component is to design and implement a mobile, AI-powered monitoring system that can continuously inspect mushroom packets, identify contamination at an early stage, and provide actionable insights to farmers. This overall objective aligns with the larger goal of introducing smart, technology-driven solutions to improve mushroom farming practices. Specifically, the main objectives are:

1. Develop a mobile camera system that moves row by row across the cultivation setup to capture images of all mushroom packets.
2. Implement a deep learning-based image recognition model to detect fungal contamination and abnormal growth patterns.
3. Provide real-time alerts and notifications to farmers via an easy-to-use dashboard.
4. Reduce manual labor and prevent the spread of contamination during inspection.

Specific Objectives

1.Capture high-quality images of mushroom packets in real cultivation environments

- Systematically capture images of all mushroom packets in operational farms using a mobile camera system that moves row by row.
- Account for real-world conditions such as changing lighting, humidity, and subtle variations in mushroom growth.
- Ensure that the deep learning model can accurately detect contamination in practical farming scenarios rather than only in controlled environments.
- Emphasize quality and variability of images, as these directly influence the model's performance.

2. Train and evaluate convolutional neural network (CNN) models for accurate detection

- Use CNN architecture to identify early signs of fungal contamination or abnormal growth patterns.
- Select the best model, train it on the collected images, and evaluate performance based on accuracy, precision, and recall.
- Focus on early detection to enable farmers to intervene before contamination spreads, reducing crop losses.
- Ensure the AI system is robust, reliable, and adaptable to different farm conditions.

3. Design a simple, intuitive interface for contamination status and feedback

- Include a user-friendly dashboard showing the contamination status of each packet or row.
- Highlight affected packets visually and allow farmers to provide feedback on the alerts.
- Enable continuous learning so the AI model improves over time based on real-world inputs.
- Ensure the interface is easy to use, even for non-technical farmers, bridging the gap between technology and practical farming.

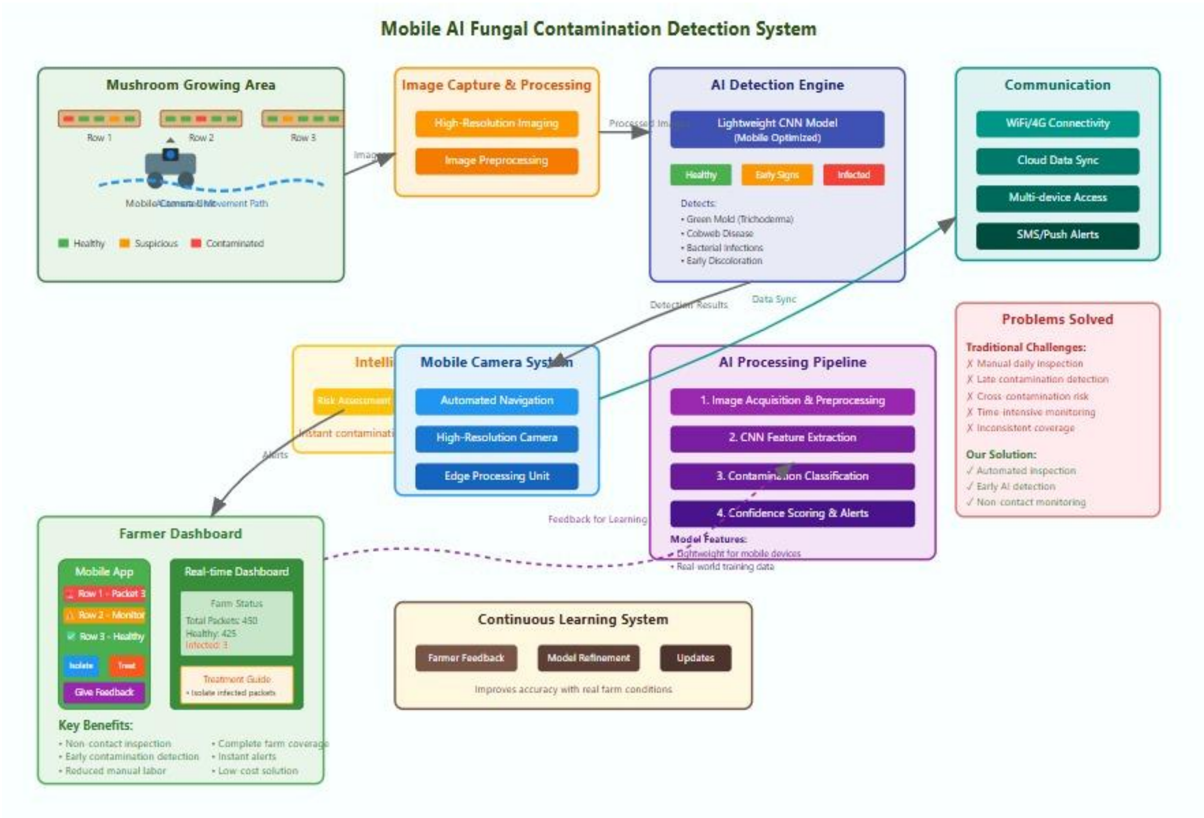
4. Ensure affordability, low power usage, and scalability for small-scale farms

- Design the hardware and software to be cost-effective and energy efficient.
- Make the system practical for farms with limited resources.
- Ensure scalability so the system can be adapted to different farm sizes and layouts.
- Deliver a solution that is not just a prototype but a viable tool for real-world deployment.

5. Demonstrate measurable improvements in early detection, crop protection, and sustainability

- Evaluate how effectively the system protects mushroom crops.
- Measure improvements in early detection rates, reduction in crop losses, and time saved compared to manual inspections.
- Demonstrate real-world benefits to validate the system’s contribution to sustainable mushroom farming.
- Support small-scale farmers in maintaining healthy crops and increasing productivity.

Methodology



The proposed system for fungal contamination detection in mushroom cultivation integrates mobile imaging and deep learning to provide a practical, real-time solution for small-scale farmers [1][2]. A mobile camera unit moves along cultivation rows, capturing high-resolution images of every mushroom packet to ensure complete coverage, overcoming the limitations of fixed cameras [3][4]. The captured images are then preprocessed through steps such as cropping, resizing, normalization, and color correction to handle variations in lighting, orientation, and background that occur under real farming conditions [5]. These preprocessed images are analyzed using a lightweight Convolutional Neural Network optimized for mobile deployment, which detects early signs of fungal contamination, such as green mold or cobweb disease, and classifies each packet as healthy, infected, or uncertain [6][7]. Detection results are displayed on a user-friendly dashboard, providing real-time visual alerts that indicate the location and severity of contamination, allowing farmers to make timely interventions [8]. Additionally, the system incorporates farmer feedback, enabling users to report false positives or missed detections, which is then used to continuously retrain and improve the model's accuracy [9]. All images, detection results, and feedback are securely stored in a local or cloud database for historical analysis, performance evaluation, and future model updates [10]. The system is evaluated under real farming conditions to ensure high detection accuracy, low latency, complete coverage of packets, and usability for farmers with minimal technical knowledge [2][5]. Its modular and scalable design allows for the integration of additional cameras, sensors, or AI models in the future, making it a low-cost, adaptable, and sustainable solution that reduces labor, minimizes contamination risks, and improves consistency in mushroom cultivation [1][3].

Project requirements

1. Functional requirements

1. **Mobile Image Capture**
    - The system must capture images of mushroom packets using a mobile camera unit that moves along cultivation rows.
    - Images should cover all mushroom packets to ensure complete monitoring.
2. **Image Preprocessing**
    - Automatically enhance, crop, and normalize captured images for consistent quality.
    - Handle variations in lighting, orientation, and background typical of real farming conditions.
3. **Fungal Contamination Detection**
    - Analyze preprocessed images using a lightweight Convolutional Neural Network (CNN) optimized for mobile deployment.
    - Detect early signs of fungal contamination such as green mold or cobweb disease.
4. **Real-Time Alerts and Notifications**
    - Provide immediate alerts to farmers via a dashboard when contamination is detected.
    - Highlight affected packets visually for easy identification and intervention.

5. **Farmer Feedback Integration**
    - Allow farmers to give feedback on detection results (e.g., false positives/negatives).
    - Use this feedback to continuously improve the AI model through incremental learning.
6. **Dashboard and Visualization**
    - Display real-time status of all mushroom packets.
    - Include visual indicators for infected, healthy, or uncertain packets.
    - Show historical detection trends and logs for farmer reference.
7. **Data Storage and Management**
    - Store captured images, detection results, and farmer feedback securely.
    - Maintain historical records for future analysis and model training.
8. **Scalability and Flexibility**
    - Support integration of additional cameras or sensors in the future.
    - Handle farms of varying sizes and layouts without performance loss.
9. **Low-Latency Processing**
    - Ensure that image analysis and alerts happen quickly enough for timely interventions.
10. **System Accessibility**
    - Dashboards should be accessible via mobile and desktop devices, enabling farmers to monitor their farms remotely.

2. Non-Functional Requirements

1. **Performance**
    - The system must process images and provide contamination detection results in real-time or near real-time to enable time interventions.
    - Image preprocessing and AI inference should have low latency, suitable for mobile deployment.

2. **Reliability & Availability**
    - The system should operate continuously during the cultivation period with minimal downtime.
    - Ensure high reliability of the camera unit and AI detection, even under variable environmental conditions.
3. **Accuracy**
    - The deep learning model should achieve high detection accuracy to minimize false positives and false negatives.
    - Regular updates and feedback integration should maintain or improve accuracy over time.
4. **Usability**
    - The dashboard and alert system must be user-friendly, intuitive, and easily accessible to farmers with minimal technical knowledge.
    - Visualization of detected contamination should be clear and actionable.
5. **Scalability**
    - The system must handle different farm sizes, additional rows, and multiple camera units without performance degradation.
    - Able to integrate more sensors or AI models in the future.
6. **Portability**
    - The mobile camera and AI system should be lightweight and portable, easily moved along cultivation rows.
    - AI models should be optimized to run efficiently on mobile devices.
7. **Security & Privacy**
    - All captured images, detection results, and farmer feedback must be securely stored.
    - Only authorized users should access farm data.
8. **Cost-Effectiveness**
    - The system should be affordable for small-scale farmers without requiring expensive hardware or infrastructure.

9. **Maintainability**

    - The system should be easy to maintain, with straightforward procedures for camera calibration, model updates, and troubleshooting.

10. **Environmental Adaptability**

    - The camera and sensors must operate reliably under real farm conditions, including varying light, humidity, and temperature.

3. User requirements

1.System Accessibility

- Users (farmers) should be able to access the dashboard on both mobile and desktop devices.
- The system should provide real-time updates on fungal contamination status.

2. Image Capture and Coverage

- The system should allow users to monitor all mushroom packets without manual inspection.
- Users should be able to move the mobile camera along rows easily and ensure complete coverage.

3.Contamination Detection

- Users need accurate detection of early fungal contamination (e.g., green mold, cobweb disease) in all mushroom packets.
- The system should minimize false positives and false negatives to avoid unnecessary interventions.

4.Alerts and Notifications

- Users should receive instant alerts when contamination is detected.
- Notifications should clearly indicate which packets are affected and the severity of contamination.

5.Visualization and Dashboard

- Users require a simple, intuitive dashboard displaying the status of all mushroom packets.
- The dashboard should allow users to track historical detection data and view trends over time.

6.Feedback Integration

- Users should be able to submit feedback on detection results to improve model accuracy.
- Feedback submission should be quick and easy on the dashboard interface.

7. Ease of Use

- The system should require minimal technical knowledge for operation.
- Users should be able to start, stop, and move the camera with ease.

8.Data Management

- Users should have access to stored images, detection results, and logs for reference.
- The system should allow users to export data if needed.

9. System Reliability

- Users require a stable and reliable system that works under typical farm conditions such as varying light, temperature, and humidity.

10. Cost-Effectiveness

- Users need an affordable solution suitable for small-scale farming without expensive hardware or software.

4. System requirements

Hardware Requirements

1. **Mobile Camera Unit**
    - Capable of capturing high-resolution images of mushroom packets.
    - Portable and lightweight to move along cultivation rows.
    - Durable to withstand farm conditions (humidity, temperature variations).
2. **Sensors (Optional for Enhancement)**
    - Environmental sensors (temperature, humidity, CO_2) for additional context if required.

3. **Processing Unit**
    - Mobile device or embedded processor capable of running lightweight CNN models.
    - Sufficient memory and storage to handle real-time image processing.
4. **Storage**
    - Local storage for captured images and detection results.
    - Cloud storage (optional) for backup and long-term data retention.
5. **Display Devices**
    - Mobile phone, tablet, or desktop to access the dashboard interface.
6. **Power Supply**
    - Battery or plug-in solution sufficient to run the camera and processing unit during operations.

Software Requirements

1. **Operating System**
    - Android, iOS, or lightweight Linux for the processing device.
2. **Image Processing & AI Software**
    - Lightweight Convolutional Neural Network (CNN) optimized for mobile deployment.
    - Image preprocessing tools for cropping, normalization, and enhancement.
3. **Dashboard / User Interface**
    - Web-based or mobile application for real-time visualization.
    - Alerts/notifications system integrated into the dashboard.
4. **Data Management**
    - Database to store images, detection results, and user feedback.
    - Secure access controls protect data privacy.

5. **Connectivity**

    - Wi-Fi, Bluetooth, or other wireless protocol for communication between the mobile camera unit and dashboard.

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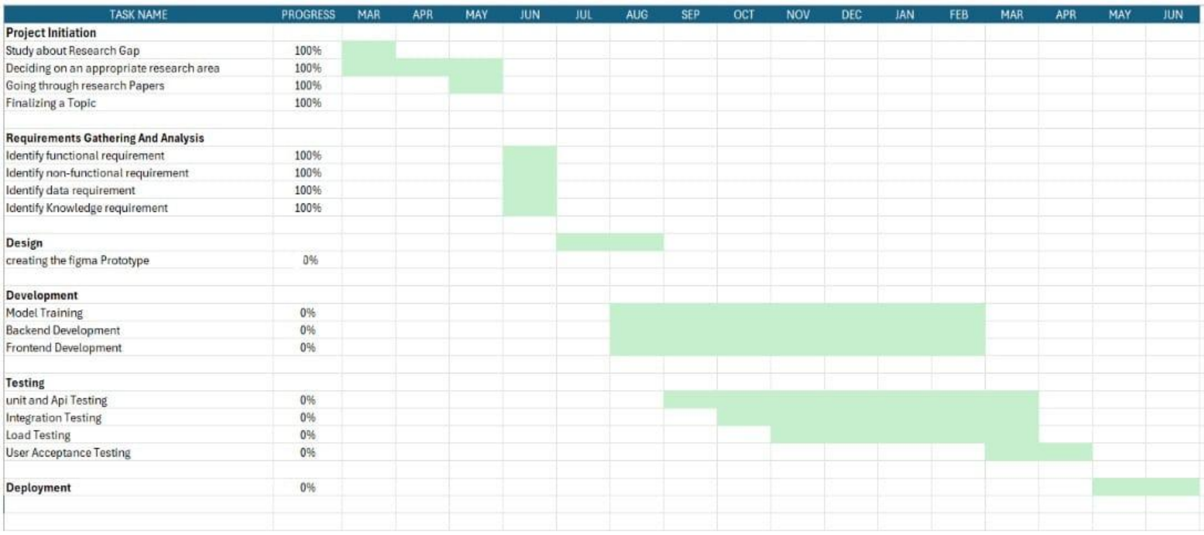
Gant Chart



TASK NAME	PROGRESS	MAR	APR	MAY	JUN	JUL	AUG	SEP	OCT	NOV	DEC	JAN	FEB	MAR	APR	MAY	JUN
Project Initiation																	
Study about Research Gap	100%																
Deciding on an appropriate research area	100%																
Going through research Papers	100%																
Finalizing a Topic	100%																
Requirements Gathering And Analysis																	
Identify functional requirement	100%																
Identify non-functional requirement	100%																
Identify data requirement	100%																
Identify Knowledge requirement	100%																
Design																	
creating the figma Prototype	0%																
Development																	
Model Training	0%																
Backend Development	0%																
Frontend Development	0%																
Testing																	
unit and Api Testing	0%																
Integration Testing	0%																
Load Testing	0%																
User Acceptance Testing	0%																
Deployment	0%																